Practical exercise 7

1. The table below shows average numbers of people (N_i) being strip-searched at an airport during a particular specified hour over the past 18 days. Consider the model $N_i \sim \text{Poisson}(\lambda_i)$, with the N_i being mutually independent and $\lambda_i = \exp(a + bi)$.

Date (i)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Total
No. of searches	11	14	15	12	11	14	15	19	21	24	21	28	30	34	39	50	60	69	487

   (a) Obtain a plot of N against i.

   (b) Fit a log-linear GLM trend model to the data. Reminder: We use the `glm` command,

   ```
   > model1=glm(N~i, family=poisson)
   ```

   Examine the output and give the equation for the fitted values of $\hat{\lambda}$.

   (c) Manually calculate the fitted value for $i = 10$. Compare with the set of values obtained using the `fitted.values` command.

   (d) Obtain a plot of residuals, using

   ```
   > plot(residuals(model1,type="response") ~ i)
   ```

   [Try `?residuals.glm` in `R` for help.]
   Construct also plots for Pearson residuals and deviance residuals. Which type of residuals does `R` plot by default, if you don't specify the type?

2. Ten groups of biological cells are exposed to different doses of a chemical. The response variable is `propdead`, the proportion of cells in the group killed, whilst the explanatory variable is `dose`, the dose of chemical administered (in suitable units).

   The results are given below:

Group	group size	number killed	propdead (y)	dose
1	72	5		1.481
2	67	11		1.675
3	57	13		1.756
4	65	22		1.866
5	58	27		1.972
6	69	36		2.017
7	71	52		2.201
8	59	46		2.253
9	60	52		2.374
10	65	65		2.523

   The data are given on Canvas in file `cells.R`. Save the file on your computer and read the data into `R` as a dataframe using a command similar to the following:

```
cells.df <- read.table("R/data/cells.R",header=T)
```

Model: proportion killed Y_i in group of size n_i is such that $n_i Y_i \sim \text{Bin}(n_i, \pi_i)$ with logit link, $\log\left(\frac{\pi_i}{1-\pi_i}\right) = b_0 + b_1 x_i$ where b_0 is the intercept and b_1 is the dose effect.

Note: The logit function is defined as $\text{logit}(p) = \log\left(\frac{\pi}{1-\pi}\right)$.

(a) Calculate the values of y_i, the proportion of cells in the group killed, and put them in a vector in `R` named `propdead`.

(b) Plot graphs of `propdead` and `logit(propdead)` against `dose`. Comment on the two graphs.

(c) Use the standard approach in `R` to fit a binomial GLM, in which we declare the proportion of cells killed as the response and include the group sizes as a set of weights (see lecture notes).

Examine the `R` output and write down an analytical expression for the equation giving the fitted value of π_i.

Compute the fitted values manually (using your expression from above) and verify their values using the `fitted()` command in `R`.

(d) Calculate the fitted value of the proportion killed for dose = 2.0, using the analytical expression from part (c). Verify your answer using the `predict()` command in `R`.

[Try `?predict.glm` for help.]